Predicting Student Reading: The Utility of Real and Nonsense Words

Kathrin E. Maki and Stephanie M. Hammerschmidt-Snidarich

UNIVERSITY OF MINNESOTA
Department of Educational Psychology

CEHD
College of Education + Human Development

Introduction

Educational accountability has led to increased use of data-based decision-making within multi-tiered systems of support (MTSS; Burns & VanDerHeyden, 2006). Research shows that interventions are more effective when correctly targeted to student skill need (Burns et al., 2008). Following seasonal screening, assessments of specific subskills such as decoding can be used to identify intervention targets (Burns & Gibbons, 2012). Decoding assessments that utilize nonsense words are often used as test items, but researchers have raised social validity concerns about such tasks, citing that reading nonsense words does not represent real learning and instructional experiences (Cunningham, 1990). The purpose of this study was to examine (a) student performance on real and nonsense words on a decoding inventory designed for the purpose of targeting instruction in the area of decoding, (b) the differential utility of real and nonsense words in predicting student reading performance, and (c) the reliability and predictive validity of a decoding inventory comprised of low frequency real words.

Method

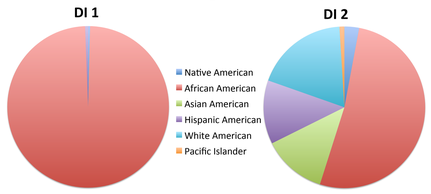
Participants

Participants attended schools with 94% (DI 1) and 82% (DI 2) free and reduced-price lunch eligibility. There were N = 54, 26 male students and N = 44, 28 female students in DI 1 and DI 2, respectively. Information on individual race/ethnicity was available only for the participants' schools. Students administered DI 1 were enrolled in a school with an average of <1% Native American, 99% African American, 0% Asian American, <1% Hispanic, 0% White, and 0% Pacific Islander students. Students administered DI 2 were enrolled in a school with an average of 3% Native American, 53% African American, 13% Asian American, 13% Hispanic, 19% White, and <1% Pacific Islander students.

Participants

	DI 1	DI 2
Grade 1	36	NA
Grade 2	29	32
Grade 3	33	22

Ethnic/Racial Percentages in Study Schools



Measures

Decoding Inventory 1. Decoding Inventory 1 was comprised of six sections with consonant-vowel-consonant (CVC) words, digraphs, blends, long vowels, r/l-controlled vowels, and diphthongs with five real and five nonsense words in each section.

Decoding Inventory 2. Decoding Inventory 2 was comprised of the same six sections as Decoding Inventory 1, but had 10 low frequency real words in each section and no nonsense words.

Oral Reading Fluency. Oral reading fluency was measured using grade-level probes from Aimsweb (2010) and following standard CBM-R procedures (Shinn, 1989). The participants read three passages for each assessment; the median value was used to evaluate student progress.

Procedures. The first round of data was collected with students in first through third grades in the participating school district during the spring of the academic school year. During spring universal benchmarking, 98 students across the three grades were randomly selected for decoding inventory 1 administration, which was administered in conjunction with the oral reading fluency (ORF) assessments. The second round of data (decoding inventory 2) was collected with students in second and third grades in the participating school district during summer school. During summer universal benchmarking, 54 students across second and third grades were administered decoding inventory 2 at two different time points to assess item-level and test-retest reliability.

Results

Decoding Inventory 1 Descriptive Statistics

	Mean	SD	Real-Nonsense Word Correlation
CVC Real Words	4.59	.91	.52
CVC Nonsense Words	3.73	1.33	
Digraphs Real Words	4.41	1.14	.58
Digraphs Nonsense Words	3.45	1.44	
Blends Real Words	3.84	1.43	.59
Blends Nonsense Words	3.15	1.59	
Long Vowel Real Words	3.77	1.43	.75
Long Vowel Nonsense Words	3.31	1.67	
R/L Controlled Real Words	3.92	1.50	.74
R/L Controlled Nonsense Words	2.92	1.58	
Dipthongs Real Words	4.04	1.40	.67
Dipthongs Nonsense Words	2.69	1.49	

	Decoding Inventory 1	Decoding Inventory 2
Coefficient Alpha	.96	.95
DI-ORF Correlation	.78	.64
Test-Retest Reliability	--	.74
Real Words-ORF Correlation	.80	--
Nonsense Words-ORF Correlation	.72	--
Real Words-DI Total Correlation	.97	--
Nonsense Words-DI Total Correlation	.96	--

Decoding Inventory 2 Item Level Analysis

Item	Item-Total Correlation
cog	.165
hen	.158
tug	.084
kin	-.17

Discussion

The results of preliminary testing on the two forms of this decoding inventory suggest that low-frequency real words are as effective for assessing decoding skills as nonsense words. Meanwhile, eliminating nonsense words from the inventory addresses concerns that such test items do not represent real learning and instructional experiences, or may lead students to attempt to read the nonsense words as a real word (e.g., read "hep" as "help").

Limitations include that DI 2 did not include first grade students, which may account for the lower correlation between the DI and an ORF measure. It is possible that re-testing the assessment with a larger sample that includes first graders may reduce possible range restriction and more accurately assess decoding ability.

Future Directions

DI 2 analyses resulted in four items with low item-total correlations suggesting that these items are not a good measure of overall student decoding. Thus, future research will analyze a new decoding inventory on which these four items have been changed to four new items. Reliability, as well as concurrent and predictive validity of the new decoding inventory, will be examined. Specifically, results of the measure will be correlated with a nationally normed measure of decoding ability and an oral reading fluency measure. A key purpose of this study was to improve the psychometric quality of a decoding inventory that can serve as a quick diagnostic tool to target decoding instruction, thus future research will examine the measure's effectiveness for this purpose. The social validity of the measure should also be formally assessed.

Decoding: Targeting Interventions

A. Short vowels in CVC words	8 / 10 use P-1 or P-2 intervention > 9 Move onto **B.**
B. Short vowels and digraphs	8 / 10 use P-3 intervention > 9 Move onto **C.**
C. Consonant blends with short vowels	8/ 10 use P-4 intervention > 9 Move onto **D.**
D. vowel spellings	8/ 10 use P-4 intervention > 9 Move onto **E.**
E. r- and l-controlled vowels	8/ 10 use P-5 intervention > 9 Move onto **F.**
F. Variant vowels and diphthongs	8/ 10 use P-5 or P-6 intervention > 9 Move onto **Fluency**

References

aimswebTM (2012). *Reading Curriculum-Based Measurement Administration and Scoring Guide.* Retrieved from http://www.aimsweb.com/

Burns, M. & Gibbons, K. (2012). *Implementing response-to-intervention in elementary and secondary schools: Procedures to assure scientific-based practices* (2nd ed.). New York: Routledge.

Burns, M.K. & VanDerHeyden, A.M (2006). Using response to intervention to assess learning disabilities: Introduction to the special series. *Assessment for Effective Intervention*, 32, 3-5.

Burns, M. K., VanDerHeyden, A. M., & Boice, C. H. (2008). Best practices in intensive academic interventions. *Best practices in school psychology V*, 1151-1162.

Cunningham, P. (1990). The Names Test: A quick assessment of decoding ability. *The Reading Teacher*, 124-129.

Shinn, M. R. (Ed.). (1989). *Curriculum-based measurement: Assessing special children*. Guilford Press.

UNIVERSITY OF MINNESOTA
Driven to Discover℠

Department of Educational Psychology
250 Education Sciences Building
56 E. River Road, Minneapolis, MN 55455

Contacts: Kathrin Maki: makix312@umn.edu
Stephanie Snidarich: snida009@umn.edu